

CODE SPORT



In this month's column, we feature a set of questions related to algorithms, data structures and operating systems.

For the past few months, we have been discussing information retrieval, Natural Language Processing (NLP) and the algorithms associated with them. We take a break from that to feature a set of computer science interview questions for our readers.

- (1) You are given a set of N strings and asked to find out if there are any duplicates among these strings. If so, find all the strings that are duplicated.
- (2) You are given a set A of N strings. You can do any amount of preprocessing on these strings. Then, you will be presented with a query string and asked to check if it has a duplicate in set A . This process will be repeated a large number of times (K), with K being much larger than N . What data structure will you use to solve this problem?
- (3) You are given a matrix of $M \times N$ integers and asked to rearrange the matrix such that each row and each column are sorted in descending order. What is the time complexity of your solution?
- (4) You are given two strings $s1$ and $s2$. You have to provide a method for converting string $s1$ to string $s2$ by means of applying the following operations any number of times: (a) insert a character, and (b) delete a character. Can you write a program to do this conversion with the minimum number of operations?
- (5) The age of each student enrolled in your class is provided in an array of N integers. You are asked to write a program that prints out how many students there are in each age category. What is the time complexity of your solution?
- (6) We are all familiar with the stack data structure, which typically supports the operations 'push' and 'pop' in $O(1)$ time complexity and exhibits 'Last In, First Out' (LIFO) behaviour. You are asked to write a special stack data

structure that supports the following two operations: Max and Second Max. These operations should also be supported with $O(1)$ time complexity.

- (7) Given an array of N integers, find the number of unordered pairs in the array. For example, if you are given the array as $\{7, 9, 8\}$, the answer is 1 because among all the pairs of elements in the array $(7, 9)$, $(9, 8)$ and $(7, 8)$, there is only one unordered pair, namely, $(9, 8)$ whereas the pairs $(7, 9)$ and $(7, 8)$ are ordered. What is the time complexity of your solution?
- (8) You are given a binary search tree of N integers. You are provided two numbers (X, Y) , and asked to return all the members of the binary search tree which lie between these two numbers, X and Y . Can you write a code snippet for the same?
- (9) You are given an integer X , which is represented as N decimal digits with each digit x being a value between 0 and 9. You are allowed to perform K swaps, where one swap can swap two consecutive digits. Write an algorithm that will maximise the value of X after K swaps. For example, if you are given the number X as 867, and $K=1$, then the maximised value is 876.
- (10) You are given an array of N integers. Some of the numbers are positive and some are negative. The number of positive numbers in the array need not equal the number of negative numbers. You are asked to rearrange the array such that a positive number is followed by a negative number, while maintaining the order of appearance of the numbers. If you run out of positive numbers or negative numbers, then the remaining class of numbers can appear at the end of the array. For example, given the array, $\{7, 8, 10, 11, -3, -5\}$, the rearranged array is $\{7, -3, 8, -5, 10, 11\}$. Can you come up with an $O(N)$ solution for the same?